Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce Department of Information Technology
(B.Sc.I.T Semester V)

Full Stack development MERN Practical

Practical– X

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Practical 10: Full Stack Execution and Validation

Aim

a. Integrate React frontend with Node and Express backend.

b. Perform insert, update, and delete operations through UI.

c. Display database records dynamically in React.

d. Implement frontend form validation.

e. Test all APIs using Postman.

f. Execute and verify the complete working of the frontend, backend, and database together.

App.jsx:-

```
import { useEffect, useState } from "react";
import axios from "axios";
const API = "http://localhost:3000/students";
function App() {
  const [students, setStudents] = useState([]);
  const [name, setName] = useState("");
  const [course, setCourse] = useState("");
```

```
const [marks, setMarks] = useState("");
const [editId, setEditId] = useState(null);
const [error, setError] = useState("");
// Get student records
const getStudents = async () => {
  try {
    const response = await axios.get(API);
    setStudents(response.data);
    setError("");
  } catch (err) {
    setError("Unable to load students. Check the backend.");
  }
};
useEffect(() => {
  getStudents();
}, []);
// Clear form
const clearForm = () => {
  setName("");
  setCourse("");
  setMarks("");
  setEditId(null);
  setError("");
};
// Add or update student
const saveStudent = async (e) => {
  e.preventDefault();
  setError("");
  if (!name.trim() || !course.trim() || marks === "")
```

```
setError("Please fill in all fields.");
      return;
    }
    if (
      !Number.isFinite(Number(marks)) ||
      Number(marks) < 0 ||
      Number(marks) > 100
    ) {
      setError("Marks must be between 0 and 100.");
      return;
    }
    const student = {
      name: name.trim(),
      course: course.trim(),
      marks: Number(marks),
    };
    try {
      let response;
      if (editId) {
        response = await axios.put(`${API}/${editId}`, student);
      } else {
        response = await axios.post(API, student);
      }
      alert(response.data.message);
      clearForm();
      await getStudents();
    } catch (err) {
      setError(
        err.response?.data?.message || "Unable to save student."
```

```
    );    }  };
  // Load details for editing
  const editStudent = (student) => {
    setEditId(student._id);
    setName(student.name);
    setCourse(student.course);
    setMarks(String(student.marks));
    setError("");
  };
  // Delete student
  const deleteStudent = async (id) => {
    if (!window.confirm("Do you want to delete this student?")) {
      return;    }
    try {
      const response = await axios.delete(`${API}/${id}`);
      alert(response.data.message);
      if (editId === id) {
        clearForm();      }
      await getStudents();
    } catch (err) {
      setError(
        err.response?.data?.message || "Unable to delete student."
      );  }  };
  return (
    <div>
      <h1>Student Management System</h1>
      <h2>{editId ? "Update Student" : "Add Student"}</h2>
      <form onSubmit={saveStudent}>
        <input
```

```
        type="text"
        placeholder="Student Name"
        value={name}
        onChange={(e) => setName(e.target.value)}
      />
      <input
        type="text"
        placeholder="Course"
        value={course}
        onChange={(e) => setCourse(e.target.value)}
      />
      <input
        type="number"
        placeholder="Marks (0-100)"
        min="0"
        max="100"
        step="any"
        value={marks}
        onChange={(e) => setMarks(e.target.value)}
      />
      <button type="submit">
        {editId ? "Update Student" : "Add Student"}
      </button>
      {editId && (
        <button type="button" onClick={clearForm}>
          Cancel
        </button>
      )}
    </form>
```

```
{error && <p style={{ color: "red" }}>{error}</p>}
<h2>Student List</h2>
<table border="1" cellPadding="8">
  <thead>
    <tr>
      <th>Name</th>
      <th>Course</th>
      <th>Marks</th>
      <th>Actions</th>
    </tr>
  </thead>
  <tbody>
    {students.length === 0 ? (
      <tr>
        <td colSpan="4">No students available.</td>
      </tr>
    ) : (
      students.map((student) => (
        <tr key={student._id}>
          <td>{student.name}</td>
          <td>{student.course}</td>
          <td>{student.marks}</td>
          <td>
            <button
              type="button"
              onClick={() => editStudent(student)}
            >
              Edit
```

```
                </button>
                <button
                  type="button"
                  onClick={() => deleteStudent(student._id)}
                >
                  Delete
                </button>
              </td>
            </tr>
          ))
        )}
      </tbody>
    </table>
  </div>  );}
export default App;
```

Server.js:-

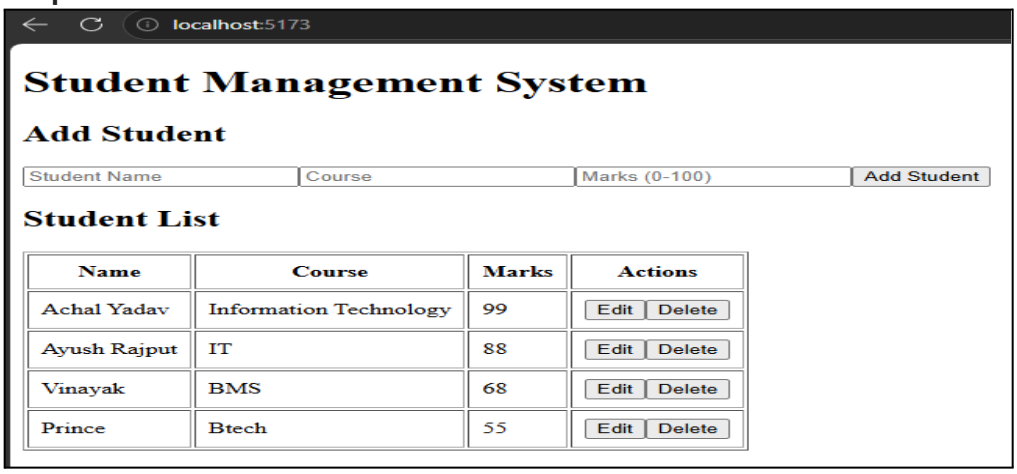
```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
app.use(cors());
app.use(express.json());
// Connect to MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
 .then(() => {
 console.log("MongoDB Connected Successfully");
 app.listen(3000, () => {
 console.log("Server running on port 3000");
 });
 })
 .catch((error) => {
 console.log("MongoDB Connection Failed:", error.message);
 });
 // Student model with validation
const Student = mongoose.model("Student", {
 name: {
 type: String,
 required: [true, "Student name is required"],
 trim: true
 },
```

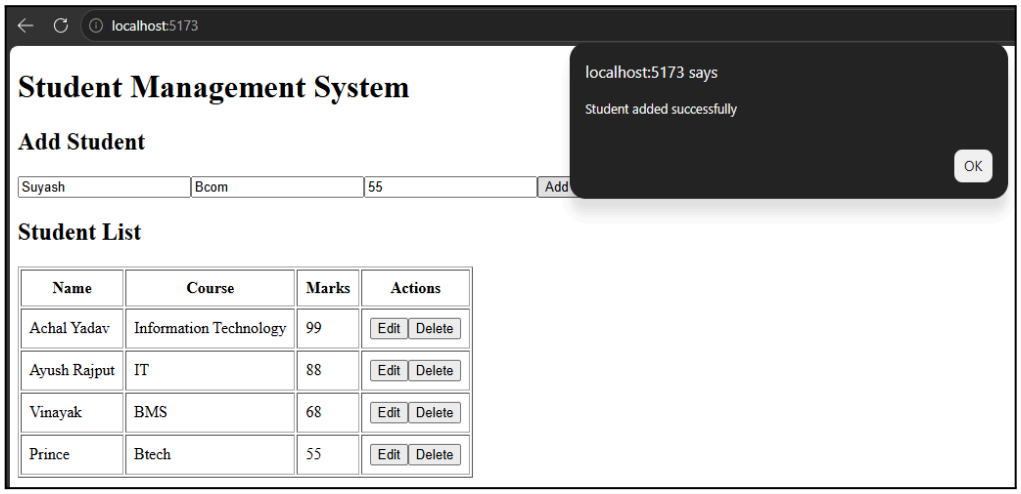
```
 course: {
 type: String,
 required: [true, "Course is required"],
 trim: true
 },
 marks: {
 type: Number,
 required: [true, "Marks are required"],
 min: [0, "Marks cannot be less than 0"],
 max: [100, "Marks cannot exceed 100"]
 }
});
// Validate student data
function validateStudent(data) {
 const { name, course, marks } = data;
 if (
 typeof name !== "string" ||
 !name.trim() ||
 typeof course !== "string" ||
 !course.trim() ||
 marks === "" ||
 marks === undefined ||
 !Number.isFinite(Number(marks)) ||
 Number(marks) < 0 ||
 Number(marks) > 100
 ) {
 return false;
 }
 return true;
}

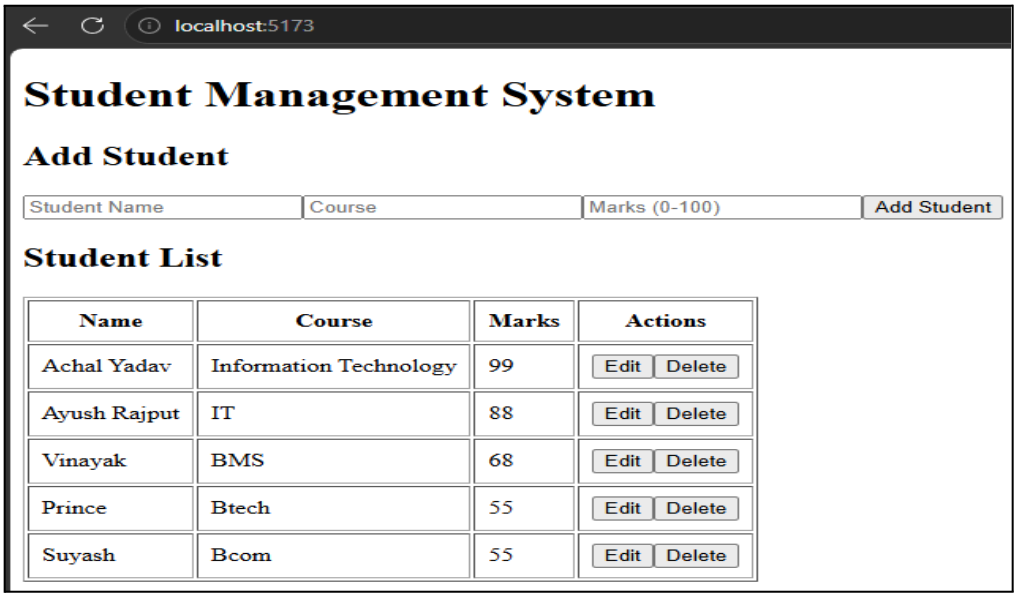
// GET: Display all students
app.get("/students", async (req, res) => {
 try {
 const students = await Student.find();
 res.json(students);
 } catch (error) {
 res.status(500).json({ message: "Error fetching students" });
 }
})
// POST: Add a student
app.post("/students", async (req, res) => {
 if (!validateStudent(req.body)) {
 return res.status(400).json({
 message: "Enter a valid name, course and marks between 0 and 100"
 });
 }
 try {
 const student = await Student.create({
 name: req.body.name.trim(),
 course: req.body.course.trim(),
 marks: Number(req.body.marks)
```

```
 });
 res.status(201).json({
 message: "Student added successfully",
 student
 });
 } catch (error) {
 res.status(400).json({ message: "Unable to add student" });
 }
});
// PUT: Update a student
app.put("/students/:id", async (req, res) => {
 if (!mongoose.isValidObjectId(req.params.id)) {
 return res.status(400).json({ message: "Invalid student ID" });
  }
 if (!validateStudent(req.body)) {
 return res.status(400).json({
 message: "Enter a valid name, course and marks between 0 and 100"
 });
 }
 try {
 const student = await Student.findByIdAndUpdate(
 req.params.id,
 {
 name: req.body.name.trim(),
 course: req.body.course.trim(),
 marks: Number(req.body.marks) },
 { new: true, runValidators: true } );
 if (!student) {
 return res.status(404).json({ message: "Student not found" }); }
 res.json({
 message: "Student updated successfully",
 student
 });
 } catch (error) {
 res.status(400).json({ message: "Unable to update student" });
 }});
// DELETE: Delete a student
app.delete("/students/:id", async (req, res) => {
 if (!mongoose.isValidObjectId(req.params.id)) {
 return res.status(400).json({ message: "Invalid student ID" });
 }
 try {
 const student = await Student.findByIdAndDelete(req.params.id);
 if (!student) {
 return res.status(404).json({ message: "Student not found" });
 }
 res.json({ message: "Student deleted successfully" });
 } catch (error) {
 res.status(500).json({ message: "Error deleting student" });
 }
});
```

Output:-


localhost:5173
Student Management System
Add Student
Student Name
Course
Marks (0-100)
Add Student
Student List
Name
Course
Marks
Actions
Achal Yadav
Information Technology
99
Edit
Delete
Ayush Rajput
IT
88
Edit
Delete
Vinayak
BMS
68
Edit
Delete
Prince
Btech
55
Edit
Delete


localhost:5173
Student Management System
localhost:5173 says
Student added successfully
OK
Add Student
Suyash
Bcom
55
Student List
Name
Course
Marks
Actions
Achal Yadav
Information Technology
99
Edit
Delete
Ayush Rajput
IT
88
Edit
Delete
Vinayak
BMS
68
Edit
Delete
Prince
Btech
55
Edit
Delete


localhost:5173
Student Management System
Add Student
Student Name
Course
Marks (0-100)
Add Student
Student List
Name
Course
Marks
Actions
Achal Yadav
Information Technology
99
Edit
Delete
Ayush Rajput
IT
88
Edit
Delete
Vinayak
BMS
68
Edit
Delete
Prince
Btech
55
Edit
Delete
Suyash
Bcom
55
Edit
Delete